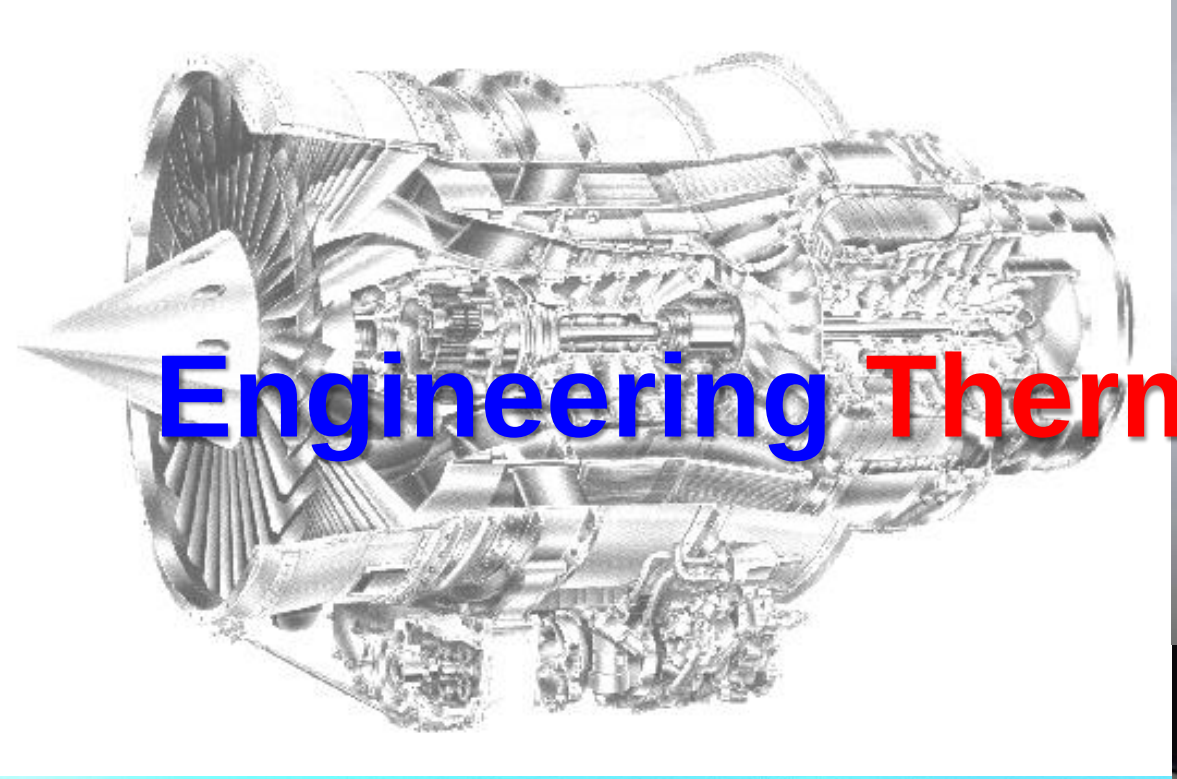
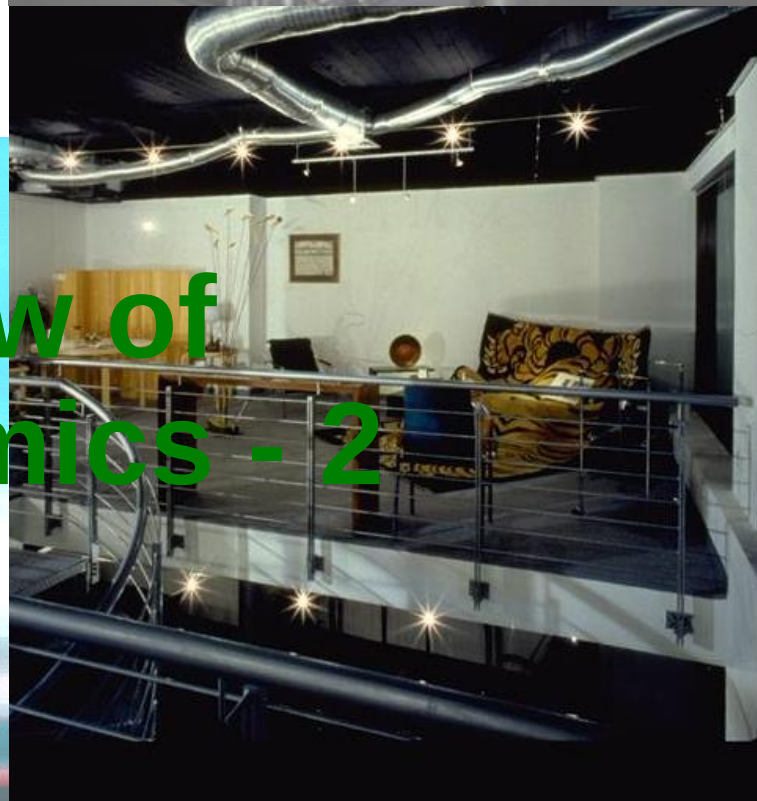


Engineering Thermodynamics



The 2nd Law of Thermodynamics - 2

Schedule

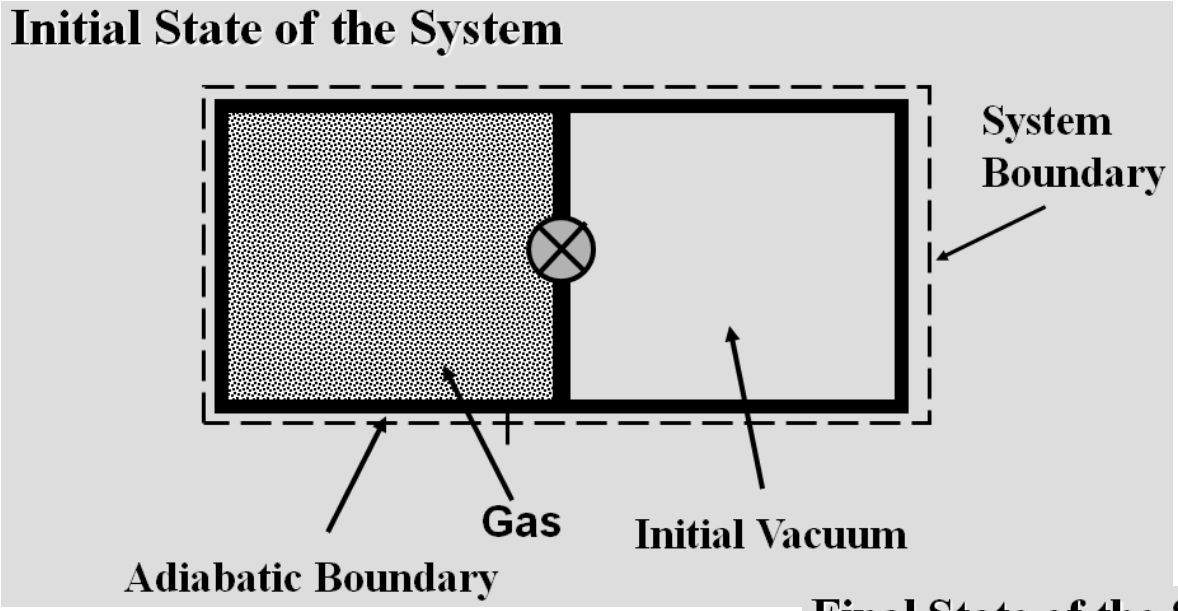
Lectures	Subjects
1	Schedule & Introduction
2	Basic concepts - 1
3	Basic concepts - 2
4	Basic concepts - 3
5	Basic concepts - 4
6	Basic concepts - 5
7	1st law of thermodynamics - 1
8	1st law of thermodynamics - 2
9	1st law of thermodynamics - 3
10	1st law of thermodynamics - 4
11	1st law of thermodynamics - 5
12	1st law of thermodynamics - 6
13	Sample Problems - 1
	Midterm exam

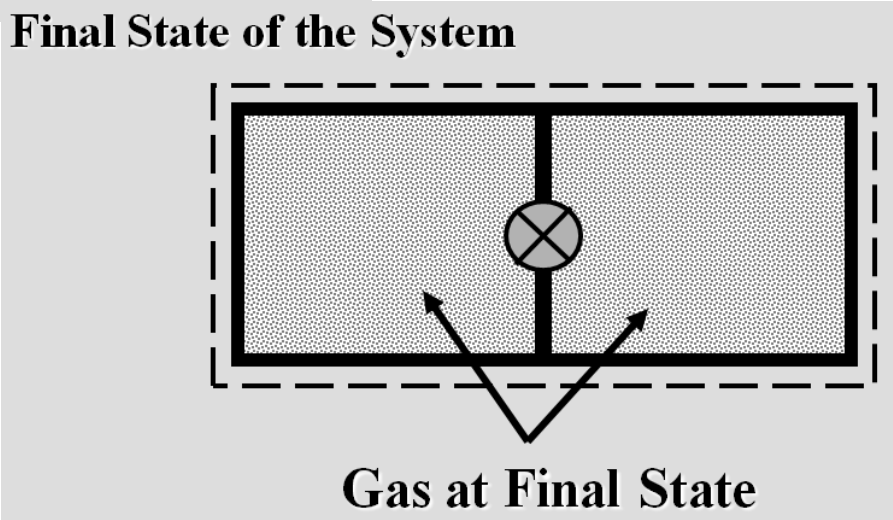
Lectures	Subjects
14	2nd law of thermodynamics - 1
15	2nd law of thermodynamics - 2
16	2nd law of thermodynamics - 3
17	2nd law of thermodynamics - 4
18	2nd law of thermodynamics - 5
19	2nd law of thermodynamics - 6
20	2nd law of thermodynamics - 7
21	2nd law of thermodynamics - 8
22	Cycles - 1
23	Cycles - 2
24	Cycles - 3
25	Cycles - 4
26	Sample Problems - 2
	Final exam

Outline

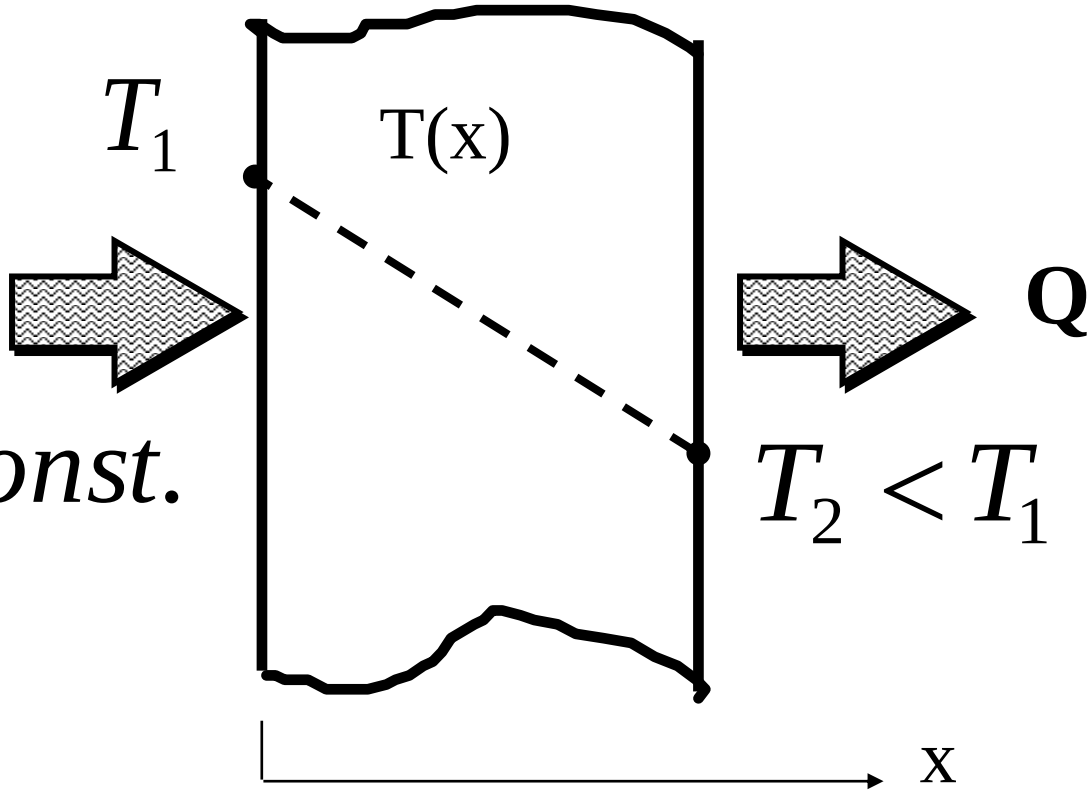
- The Second Law of Thermodynamics
  - Irreversibility
  - Reversible cycle
  - Carnot cycle

Some causes of irreversible processes
Example - Free expansion



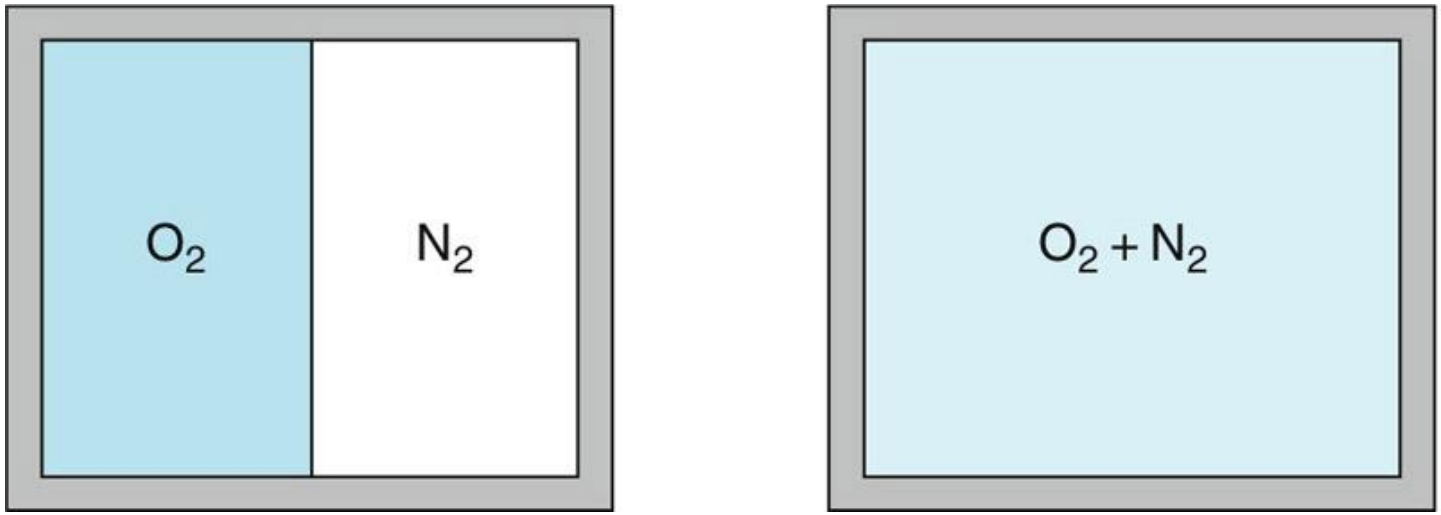


Example - Heat transfer over a finite temperature difference

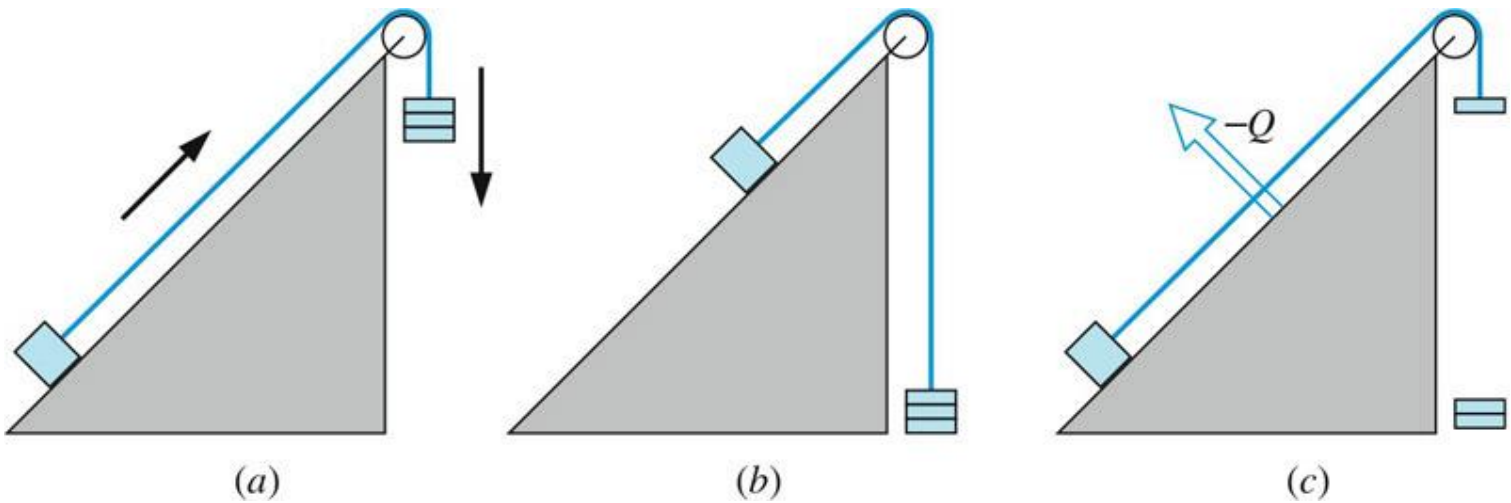


$$\dot{Q}_{1-2} = const.$$

Example - Spontaneous mass transfer via mixing

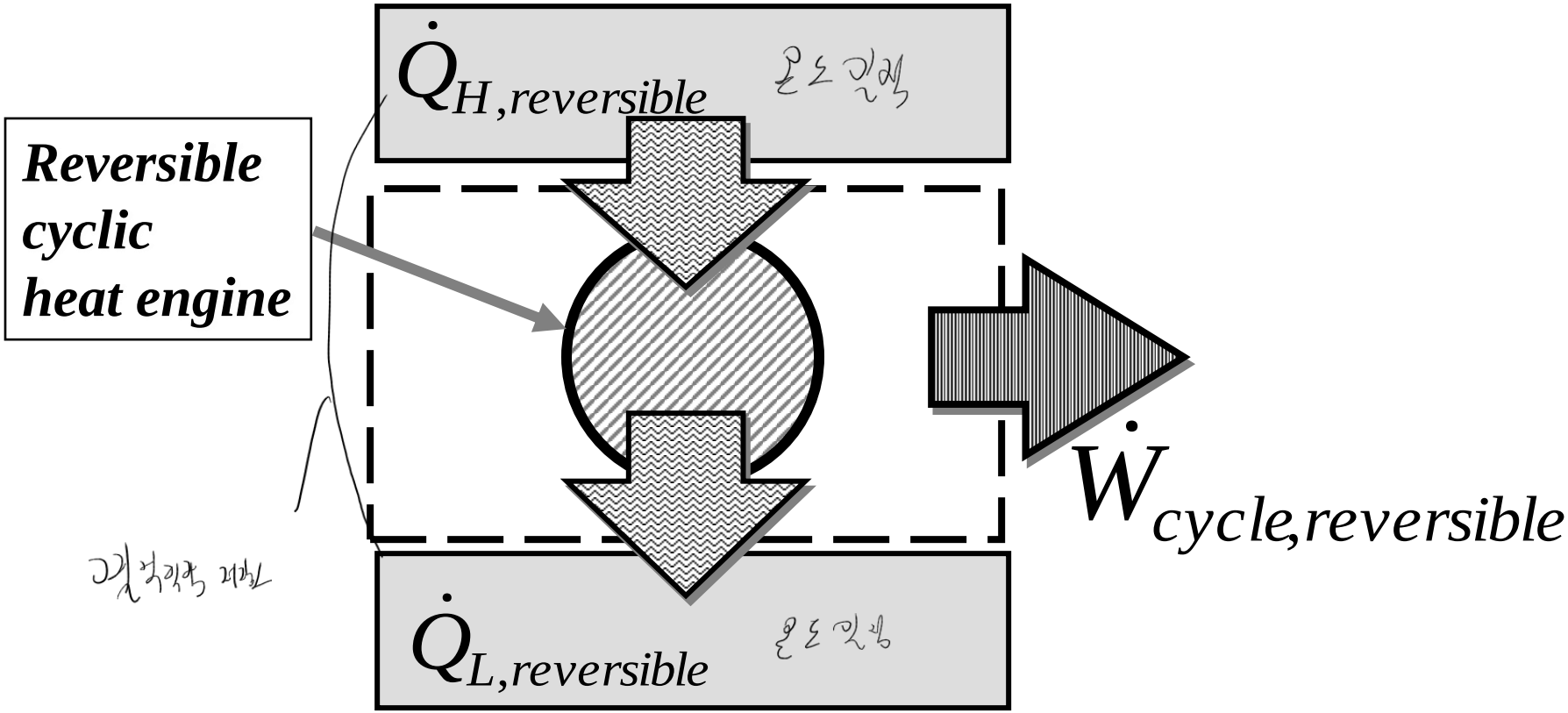


Example - Heat generation due to friction



REVERSIBLE CYCLES

A reversible power cycle is one comprising a sequence of reversible processes:



Performance of power cycles and the Second Law

- The Second Law mandates two thermal reservoirs.
- The Second Law mandates a temperature difference in order to operate the cycle.
  - *Applies to work producing and work consuming cycles.*

Performance of power cycles

- **First Law efficiency for work producing cycles**

1법칙

$$\eta = \frac{W_{cycle}}{Q_H} = \frac{Q_H - Q_C}{Q_H} = 1 - \frac{Q_c}{Q_H}$$

Performance of power cycles

- **Coefficient of performance for working consuming cycles: *β* or *COP***

$$COP = \frac{\dot{Q}_{desired}}{\dot{W}_{cycle}}$$

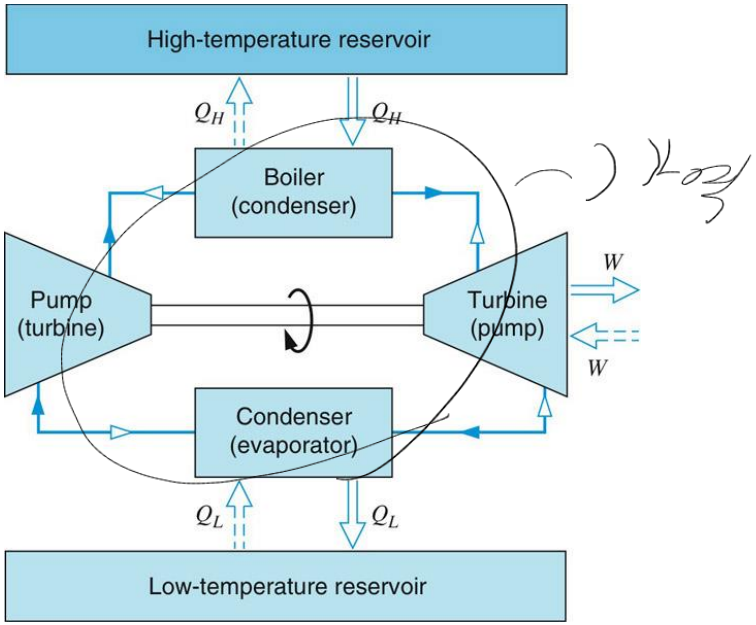
$$COP_{refrigeraiton} = \frac{\dot{Q}_C}{\dot{W}_{cycle}}, \quad COP_{heat-pump} = \frac{\dot{Q}_H}{\dot{W}_{cycle}}$$

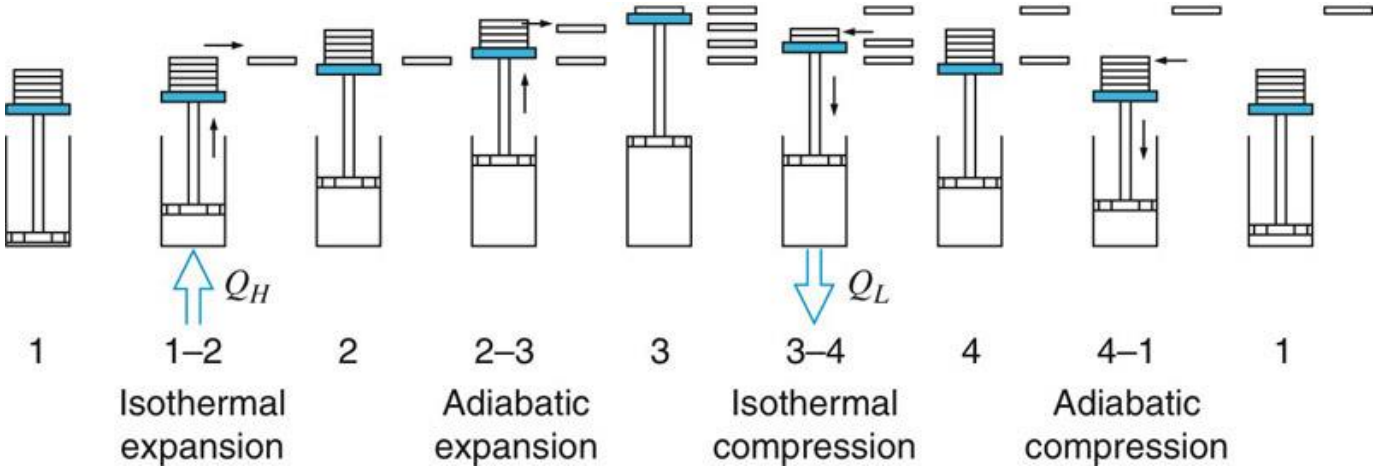
The Carnot cycle

A fully reversible power cycle

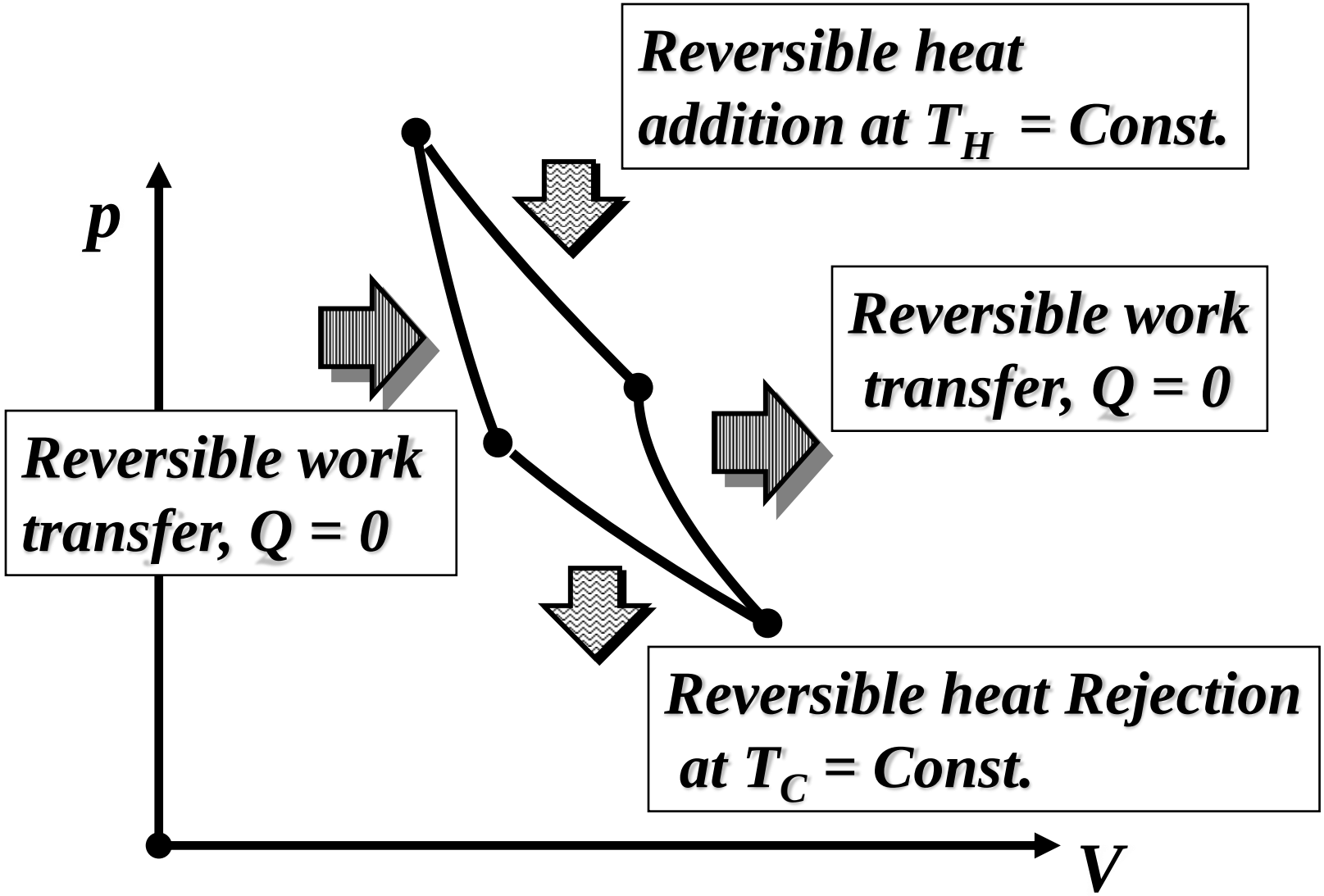
Carnot cycle

- The *Carnot cycle* is a reversible cycle that is composed of four reversible processes, two isothermal and two adiabatic.



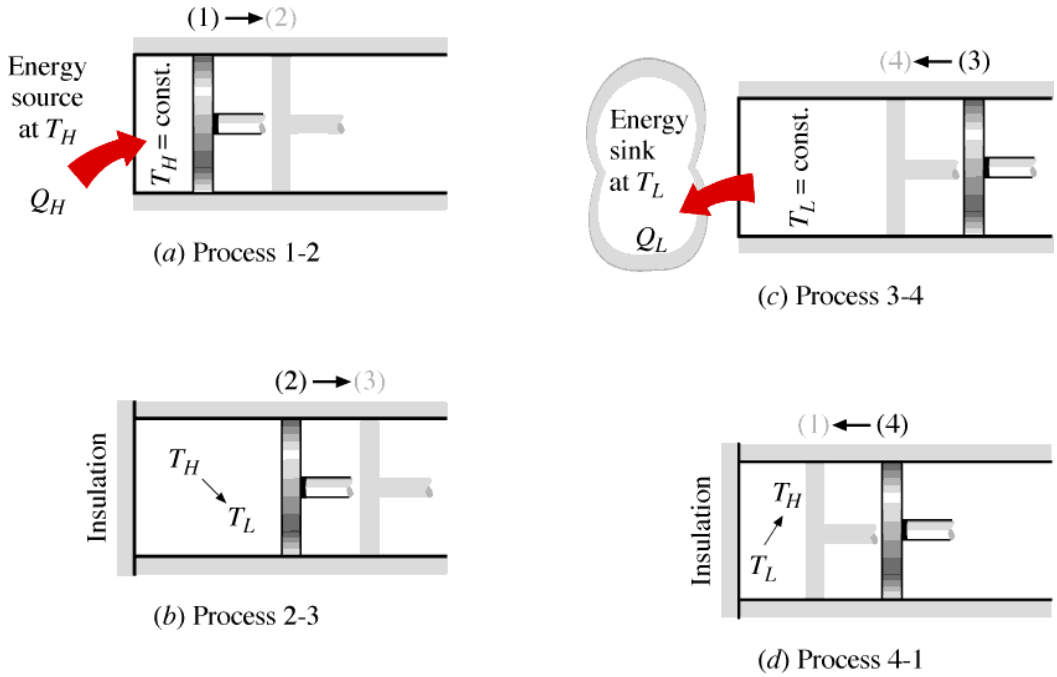


Carnot cycle



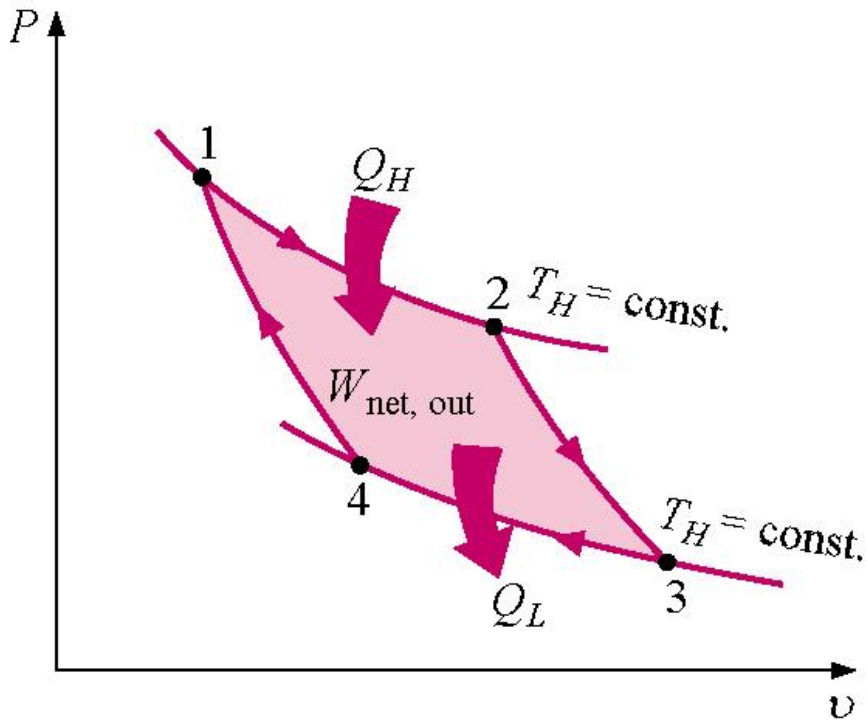
Carnot cycle

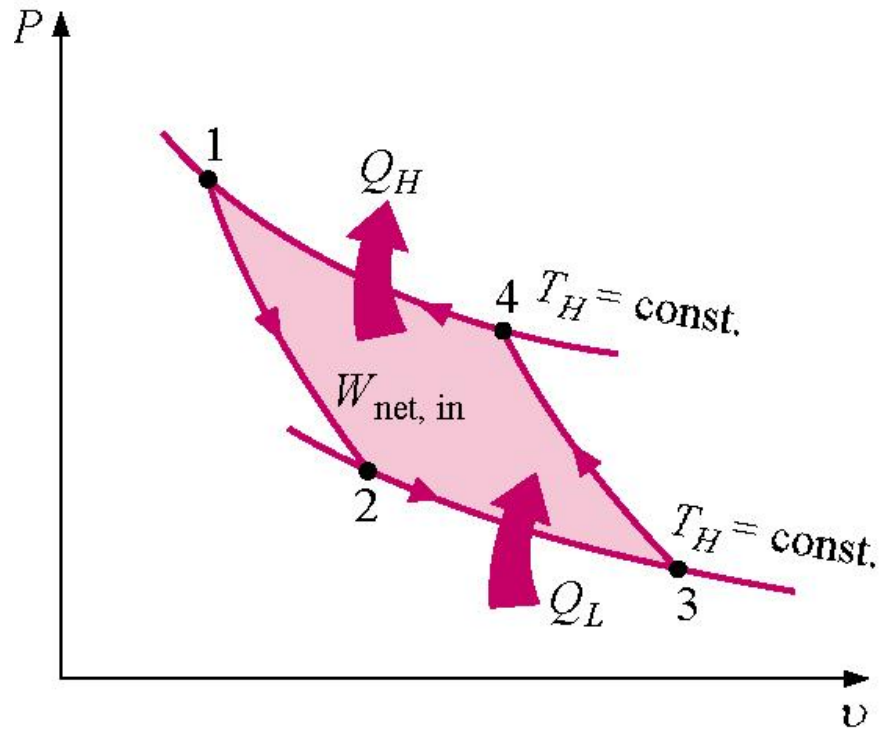
- **Can be either a work producing cycle or a work consuming cycle.**
- ***Can be operated in either direction.***
- **Represents an idealization of most power cycles (work producing and work consuming) *because of its reversibility and generality.***



Execution of the Carnot cycle in a closed system

P-v Diagram of the Carnot Cycle and the Reversed Carnot Cycle



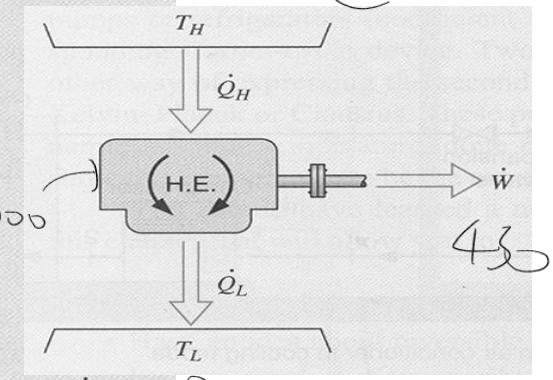


Example

Example 1:

Let us consider the heat engine, shown schematically in Fig. 7.25, that receives a heat-transfer rate of 1 MW at a high temperature of 550°C and rejects energy to the ambient surroundings at 300 K. Work is produced at a rate of 450 kW. We would like to know how much energy is discarded to the ambient surroundings and the engine efficiency and compare both of these to a Carnot heat engine operating between the same two reservoirs.

550°C



1000

450

300

Solution

If we take the heat engine as a control volume, the energy equation gives

$$\dot{Q}_L = \dot{Q}_H - \dot{W} = 1000 - 450 = 550 \text{ kW}$$

and from the definition of the efficiency

$$\eta_{\text{thermal}} = \dot{W}/\dot{Q}_H = 450/1000 = 0.45$$

For the Carnot heat engine, the efficiency is given by the temperature of the reservoirs

$$\eta_{\text{Carnot}} = 1 - \frac{T_L}{T_H} = 1 - \frac{300}{550 + 273} = 0.635$$

The rates of work and heat rejection become

$$\dot{W} = \eta_{\text{Carnot}}\dot{Q}_H = 0.635 \times 1000 = 635 \text{ kW}$$

$$\dot{Q}_L = \dot{Q}_H - \dot{W} = 1000 - 635 = 365 \text{ kW}$$

The actual heat engine thus has a lower efficiency than the Carnot (ideal) heat engine, with a value of 45% typical for a modern steam power plant. This also implies that the actual engine rejects a larger amount of energy to the ambient surroundings (55%) compared with the Carnot heat engine (36%).